# MSG No. 218.82 $P\bar{4}3n1'$ [ Type II, cubic ]

Table 1: Wyckoff site: 2a, site symmetry: 23.1'

| No. | position | mapping |
|---|---|---|
| 1 | [0, 0, 0] | [1,2,3,4,5,6,7,8,9,10,11,12,25,26,27,28,29,30,31,32,33,34,35,36] |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | [13,14,15,16,17,18,19,20,21,22,23,24,37,38,39,40,41,42,43,44,45,46,47,48] |

Table 2: Wyckoff site: 6b, site symmetry: 222..1'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | [1,2,3,4,25,26,27,28] |
| 2 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | [5,8,10,12,29,32,34,36] |
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | [6,7,9,11,30,31,33,35] |
| 4 | $\left[\frac{1}{2}, 0, 0\right]$ | [13,14,21,22,37,38,45,46] |
| 5 | $\left[0, 0, \frac{1}{2}\right]$ | [15,16,23,24,39,40,47,48] |
| 6 | $\left[0, \frac{1}{2}, 0\right]$ | [17,18,19,20,41,42,43,44] |

Table 3: Wyckoff site: 6c, site symmetry: -4..1'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{2}, 0\right]$ | [1,2,13,14,25,26,37,38] |
| 2 | $\left[\frac{3}{4}, \frac{1}{2}, 0\right]$ | [3,4,21,22,27,28,45,46] |
| 3 | $\left[0, \frac{1}{4}, \frac{1}{2}\right]$ | [5,12,17,19,29,36,41,43] |
| 4 | $\left[\frac{1}{2}, 0, \frac{1}{4}\right]$ | [6,9,16,23,30,33,40,47] |
| 5 | $\left[\frac{1}{2}, 0, \frac{3}{4}\right]$ | [7,11,15,24,31,35,39,48] |
| 6 | $\left[0, \frac{3}{4}, \frac{1}{2}\right]$ | [8,10,18,20,32,34,42,44] |

Table 4: Wyckoff site: 6d, site symmetry: -4..1'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, 0, \frac{1}{2}\right]$ | [1,2,13,14,25,26,37,38] |
| 2 | $\left[\frac{3}{4}, 0, \frac{1}{2}\right]$ | [3,4,21,22,27,28,45,46] |
| 3 | $\left[\frac{1}{2}, \frac{1}{4}, 0\right]$ | [5,12,17,19,29,36,41,43] |
| 4 | $\left[0, \frac{1}{2}, \frac{1}{4}\right]$ | [6,9,16,23,30,33,40,47] |
| 5 | $\left[0, \frac{1}{2}, \frac{3}{4}\right]$ | [7,11,15,24,31,35,39,48] |
| 6 | $\left[\frac{1}{2}, \frac{3}{4}, 0\right]$ | [8,10,18,20,32,34,42,44] |

Table 5: Wyckoff site: 8e, site symmetry: .3.1'

| No. | position | mapping |
|---|---|---|
| 1 | $[x, x, x]$ | [1,5,6,25,29,30] |
| 2 | $[x, -x, -x]$ | [2,10,11,26,34,35] |
| 3 | $[-x, x, -x]$ | [3,7,12,27,31,36] |
| 4 | $[-x, -x, x]$ | [4,8,9,28,32,33] |

*continued ...*

Table 5

| No. | position | mapping |
|---|---|---|
| 5 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$ | [13,18,23,37,42,47] |
| 6 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$ | [14,15,19,38,39,43] |
| 7 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$ | [16,17,21,40,41,45] |
| 8 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$ | [20,22,24,44,46,48] |

Table 6: Wyckoff site: 12f, site symmetry: 2..1'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ 0]$ | [1,2,25,26] |
| 2 | $[-x,\ 0,\ 0]$ | [3,4,27,28] |
| 3 | $[0,\ x,\ 0]$ | [5,12,29,36] |
| 4 | $[0,\ 0,\ x]$ | [6,9,30,33] |
| 5 | $[0,\ 0,\ -x]$ | [7,11,31,35] |
| 6 | $[0,\ -x,\ 0]$ | [8,10,32,34] |
| 7 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | [13,14,37,38] |
| 8 | $\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$ | [15,24,39,48] |
| 9 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$ | [16,23,40,47] |
| 10 | $\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | [17,19,41,43] |
| 11 | $\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | [18,20,42,44] |
| 12 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | [21,22,45,46] |

Table 7: Wyckoff site: 12g, site symmetry: 2..1'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{2},\ 0\right]$ | [1,2,25,26] |
| 2 | $\left[-x,\ \frac{1}{2},\ 0\right]$ | [3,4,27,28] |
| 3 | $\left[0,\ x,\ \frac{1}{2}\right]$ | [5,12,29,36] |
| 4 | $\left[\frac{1}{2},\ 0,\ x\right]$ | [6,9,30,33] |
| 5 | $\left[\frac{1}{2},\ 0,\ -x\right]$ | [7,11,31,35] |
| 6 | $\left[0,\ -x,\ \frac{1}{2}\right]$ | [8,10,32,34] |
| 7 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$ | [13,14,37,38] |
| 8 | $\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$ | [15,24,39,48] |
| 9 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$ | [16,23,40,47] |
| 10 | $\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | [17,19,41,43] |
| 11 | $\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | [18,20,42,44] |
| 12 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | [21,22,45,46] |

Table 8: Wyckoff site: 12h, site symmetry: 2..1'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 0,\ \frac{1}{2}\right]$ | [1,2,25,26] |
| 2 | $\left[-x,\ 0,\ \frac{1}{2}\right]$ | [3,4,27,28] |
| 3 | $\left[\frac{1}{2},\ x,\ 0\right]$ | [5,12,29,36] |

*continued ...*

Table 8

| No. | position | mapping |
|---|---|---|
| 4 | $\left[0,\ \frac{1}{2},\ x\right]$ | `[6,9,30,33]` |
| 5 | $\left[0,\ \frac{1}{2},\ -x\right]$ | `[7,11,31,35]` |
| 6 | $\left[\frac{1}{2},\ -x,\ 0\right]$ | `[8,10,32,34]` |
| 7 | $\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$ | `[13,14,37,38]` |
| 8 | $\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$ | `[15,24,39,48]` |
| 9 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$ | `[16,23,40,47]` |
| 10 | $\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$ | `[17,19,41,43]` |
| 11 | $\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$ | `[18,20,42,44]` |
| 12 | $\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$ | `[21,22,45,46]` |

Table 9: Wyckoff site: `24i`, site symmetry: `11'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1,25]` |
| 2 | $[x,\ -y,\ -z]$ | `[2,26]` |
| 3 | $[-x,\ y,\ -z]$ | `[3,27]` |
| 4 | $[-x,\ -y,\ z]$ | `[4,28]` |
| 5 | $[z,\ x,\ y]$ | `[5,29]` |
| 6 | $[y,\ z,\ x]$ | `[6,30]` |
| 7 | $[-y,\ z,\ -x]$ | `[7,31]` |
| 8 | $[-z,\ -x,\ y]$ | `[8,32]` |
| 9 | $[-y,\ -z,\ x]$ | `[9,33]` |
| 10 | $[z,\ -x,\ -y]$ | `[10,34]` |
| 11 | $[y,\ -z,\ -x]$ | `[11,35]` |
| 12 | $[-z,\ x,\ -y]$ | `[12,36]` |
| 13 | $\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$ | `[13,37]` |
| 14 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$ | `[14,38]` |
| 15 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$ | `[15,39]` |
| 16 | $\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$ | `[16,40]` |
| 17 | $\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | `[17,41]` |
| 18 | $\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | `[18,42]` |
| 19 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | `[19,43]` |
| 20 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[20,44]` |
| 21 | $\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$ | `[21,45]` |
| 22 | $\left[x+\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$ | `[22,46]` |
| 23 | $\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ \frac{1}{2}-x\right]$ | `[23,47]` |
| 24 | $\left[z+\frac{1}{2},\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$ | `[24,48]` |